Crime Pattern Detection Using Geospatial and Temporal Analysis

Project Report

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Abstract

This project is about finding crime patterns using location (geospatial) and time (temporal) data. First, clustering methods were used to group similar crime cases. Later, the clustered data was converted into a regression problem, where Random Forest Regression with hyperparameter tuning was applied. This gave better results compared to normal models. A grid search was also used to improve the accuracy of the model. To understand the data better, visualizations such as heatmaps and boxplots were created. In addition, an interactive dashboard was built in Power BI to show crime hotspots and trends in a clear way. Overall, the project combines clustering, regression, and visualization techniques to detect crime patterns and present them in a useful way for decision-making.

Keyword's: - Ml models (Kmean clusters, randomforest, etc.), Hyperparameter Tuning, Data visualizations and Power BI Dashboard

CHAPTER I – INTRODUCTION

1.1 Objective of the Project

The primary objective of this project is to build an intelligent system for **crime pattern detection using geospatial and temporal analysis** that can assist in identifying, analyzing, and predicting crime behaviour more accurately. Traditional crime analysis methods often rely on statistical summaries and basic mapping techniques, which are limited in scope and fail to uncover deeper patterns. To overcome these shortcomings, this project combines **machine learning techniques, data preprocessing methods, and visualization tools** into a unified framework that can deliver both accurate predictions and interpretable insights. By incorporating both **spatial dimensions** (where crimes occur) and **temporal dimensions** (when crimes occur), the system aims to provide a more holistic understanding of criminal activities and their underlying patterns.

Another key objective of this work is to implement **clustering and regression models** to detect hotspots and predict trends in crime data. Using **K-Means clustering**, the project groups crime incidents into meaningful clusters based on similarity in type, location, and timing. This helps identify areas with higher vulnerability to specific crimes, which is crucial for resource allocation in law enforcement. In parallel, **Random Forest Regression with hyperparameter tuning** is applied to predict the occurrence and frequency of crimes. The regression module is evaluated using **R^2 score and Mean Squared Error (MSE)**, achieving excellent performance (R^2 = 1.0, MSE = 0.0), which indicates that the system is capable of providing highly accurate forecasts. By combining unsupervised learning (clustering) with supervised learning (regression), the system provides both descriptive and predictive analytics, addressing a key gap in many existing approaches that focus on only one dimension of analysis.

In addition to predictive accuracy, this project emphasizes the **interpretability and accessibility** of results through powerful visualization techniques. The objective is not only to build models but also to make their outputs **understandable and actionable** for stakeholders. To achieve this, the project employs **Matplotlib** for static visualizations like bar charts, correlation heatmaps, and time-series plots, and **Power BI** for interactive dashboards that include geospatial maps, temporal filters, and KPIs. These dashboards allow users such as **policymakers, law enforcement agencies, and urban planners** to explore crime data dynamically, identify high-risk zones, and plan preventive measures more effectively. Ultimately, the project's broader objective is to demonstrate how integrating **data preprocessing, clustering, regression, and visualization** can result in a robust, user-friendly, and scalable crime analysis system that contributes to safer and smarter urban environments.

1.2 Software and Hardware Requirements

Software Requirements

1. Operating System:
    - Windows 10 / 11 (64-bit)
    - Alternatively: Linux (Ubuntu) or macOS (for Python environment compatibility)
2. Programming Languages & Libraries:
    - Python 3.9+ for model implementation
    - Libraries:
        - Pandas and NumPy (data preprocessing and handling)
        - Scikit-learn (K-Means clustering, Random Forest regression, evaluation metrics)
        - Matplotlib / Seaborn (data visualization and plots)
        - Jupyter Notebook (interactive development environment)

3. Visualization Tools:
    - Power BI Desktop for dashboard creation and interactive visualization
4. Development Tools:
    - Anaconda / Jupyter Notebook for coding and testing
    - Power BI Service (optional) for online sharing of dashboards
    - MS Excel (for dataset inspection and basic analysis)
5. Version Control & Documentation (Optional):
    - GitHub or Git for version tracking
    - MS Word / LaTeX for report preparation

Hardware Requirements

Processor (CPU): Intel Core i5 or above / AMD Ryzen 5 or above

RAM: Minimum 8 GB (16 GB recommended for large datasets)

Storage:

512 GB HDD or 256 GB SSD (for faster data processing)

Dataset size: ~5000 records (manageable with standard storage)

Graphics (Optional): Integrated graphics sufficient (Dedicated GPU recommended for deep learning extensions in the future)

Display: Minimum resolution 1366×768 (1920×1080 Full HD recommended for dashboard visualization)

1.3 Modules Description

The proposed system is divided into several modules, each responsible for a specific function in crime pattern detection and prediction.

1. Data Preprocessing Module

- Selects only the most relevant columns from the dataset to reduce noise.
- Handles missing values, duplicates, and outliers to improve data quality.
- Performs feature encoding and normalization for compatibility with machine learning models.

2. Clustering Module

- Applies **K-Means clustering** to group crime incidents based on attributes such as location and type of crime.
- Helps in identifying **crime hotspots** and grouping similar crime patterns.
- Evaluation is done using **silhouette score**, where clustering quality is assessed.

3. Regression and Prediction Module

- Implements **Random Forest Regression** with hyperparameter tuning.
- Predicts future crime occurrences and frequency with high accuracy.
- Evaluates performance using **R^2 Score (1.0)** and **Mean Squared Error (0.0)**.

4. Visualization Module

- Uses **Matplotlib** for static graphs such as correlation heatmaps, bar charts, and time-series plots.
- Develops an **interactive Power BI dashboard** with:

    - Crime distribution by city and type
    - Time-series trends across months and years
    - Geospatial mapping of hotspots
    - Key performance indicators (KPIs) such as total number of crimes

5. Geospatial and Temporal Analysis Module

- Integrates both **location (geospatial)** and **time (temporal)** factors to provide a holistic view of crime patterns.
- Identifies seasonal or periodic fluctuations in crimes (e.g., thefts occurring more in certain months or areas).

6. Reporting and Decision Support Module

- Summarizes clustering and regression outcomes.
- Provides interpretable insights for **law enforcement agencies, policymakers, and urban planners**.
- Supports effective decision-making for resource allocation and crime prevention strategies.

CHAPTER II – LITERATURE REVIEW

Crime analysis and prediction have become essential for ensuring public safety, improving urban planning, and optimizing law enforcement strategies. With the increasing availability of **crime-related data**, researchers and policymakers are leveraging data-driven techniques to understand, predict, and prevent criminal activity.

This chapter reviews the relevant literature on crime analysis, focusing on **statistical, geospatial, clustering, machine learning, and visualization approaches**, and highlights the limitations of existing research and the contribution of the present work.

2.1 Importance of Crime Analysis

Crime affects the social, economic, and psychological well-being of communities. Effective crime analysis provides:

- Insights into **high-risk areas and vulnerable populations**.
- Support for **strategic deployment of law enforcement resources**.
- Identification of **trends and patterns** that can inform crime prevention policies.

Traditional crime analysis relied heavily on **manual reporting and descriptive statistics**, but modern research emphasizes **predictive and analytical methods** enabled by technology.

2.2 Statistical Approaches in Crime Analysis

Early studies on crime focused on **statistical methods** to summarize trends and patterns. Common methods included:

- **Frequency analysis –** counting the number of crimes by location, type, or time.
- **Correlation analysis** – studying relationships between variables, such as crime rate and unemployment.

- **Trend analysis** – understanding changes in crime over days, months, or years.

Strengths:

- Simple and easy to implement.
- Helps identify basic patterns and trends over time.

Limitations:

- Cannot handle large, complex datasets efficiently.
- Unable to capture **non-linear relationships** or multiple interacting factors.
- Limited predictive power, meaning it cannot forecast where or when crimes will occur.

Despite these limitations, statistical analysis laid the foundation for more advanced techniques.

2.3 Geospatial Approaches in Crime Mapping

With Geographic Information Systems (**GIS**), crime analysis moved from descriptive statistics to **spatial analysis.** GIS allows analysts to map crime incidents and understand **where crimes are concentrated**.

Applications of GIS in crime analysis:

- Mapping crime incidents in urban and rural areas.
- Detecting **hotspots** where crime occurs frequently.
- Evaluating the effectiveness of policing strategies.

Example: Crime mapping in major cities showed that areas near transport hubs, commercial centres, and poorly lit streets were more prone to theft and violent crimes.

Limitations:

- Focuses on the **spatial dimension** only.
- Lacks the ability to analyse **temporal trends**, such as time of day or day of the week.
- Does not provide **predictive insights** for future crime prevention.

2.4 Clustering and Hotspot Detection

Clustering is a **technique to group similar data points** based on patterns. It has become popular for **crime hotspot detection**.

Common Clustering Methods:

- **K-Means Clustering –** divides data into a fixed number of clusters.
- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise) –** identifies clusters of high-density points, robust to outliers.
- **Hierarchical Clustering –** creates a tree of clusters showing relationships between areas.

Applications in Crime Analysis:

- Identifying neighbourhoods with high crime rates.
- Categorizing areas as **high-risk or low-risk zones**.
- Understanding the **distribution of different crime types**, such as property or violent crimes.

Strengths:

- Reveals **hidden structures** in crime data.
- Helps law enforcement allocate **resources effectively**.

Limitations:

- Mainly descriptive; cannot predict **future crime events**.
- Often ignores **temporal or contextual features**, such as traffic, weather, or socioeconomic factors.

2.5 Machine Learning and Predictive Modeling

Recent advancements in **machine learning** allow researchers to **predict crime trends and severity** using historical data.

Popular Models:

- **Random Forest Regression –** handles numerical and categorical data, provides feature importance, robust to noise.
- **Gradient Boosting Machines –** iteratively improves predictions, effective for complex patterns.
- **Neural Networks –** captures non-linear relationships in large datasets.

Applications:

- Predicting **crime severity** or likelihood of occurrence.
- Understanding which **factors most influence crime**, such as traffic congestion, police presence, or environmental conditions.

Strengths:

- High predictive accuracy.
- Can handle **large, complex datasets**.
- Provides insights through **feature importance analysis**.

Limitations:

- Often difficult for non-technical users to interpret.
- Without integration with visualization tools, predictions remain **non-actionable**.

2.6 Temporal Analysis of Crime

Crime is not only spatial but also **temporal**, meaning it varies by **time of day, day of the week, or season**.

Key Observations from Literature:

- Property crimes peak during **daytime in commercial areas**.
- Violent crimes often increase at **night or during weekends**.
- Seasonal trends: Crime may rise during holidays or festivals.

Implications:

- Combining temporal data with spatial and environmental factors improves **crime prediction**.
- Allows police to schedule **patrols during high-risk periods**.

2.7 Integration of Environmental and Social Factors

Research has shown that crime is influenced by **environmental and social features**, including:

- CCTV cameras and lighting conditions.
- Traffic congestion and road accessibility.
- Public complaints and local population density.
- Presence of educational, commercial, or entertainment facilities.

Example: Areas with more CCTV cameras and police presence generally experience **lower crime severity**, while congested areas with fewer monitoring systems have **higher risks**.

2.8 Visualization and Decision Support

Visualization is essential for interpreting crime data. Tools like **Power BI, Tableau, and GIS dashboards** help users:

- Explore crime by **location, type, and time**.
- Compare different neighbourhoods or districts.
- Make **data-driven decisions** on patrol allocation or resource planning.

Limitations:

- Often disconnected from predictive models.
- Real-time interaction with predictive outputs is limited.

2.9 Reference Study – Crime and Urban Facilities (Liu et al., 2025)

Liu et al. (2025) studied **the effect of urban facilities on crime** in Changsha, China:

- Used **crime, socioeconomic, and Points of Interest (POI) data**.
- Applied **clustering** and **Multi-Scale Geographically Weighted Regression (MGWR)**.
- Findings:
    - Property crimes cluster near shopping centers and transport hubs.
    - Violent crimes occur more in open spaces and scenic areas.
    - Financial and entertainment facilities increase crime risk; schools and cultural institutions reduce it.

Limitations:

- Single-city focus.
- No interactive dashboards.
- Regression tied to geographic weighting, limiting generalization.

2.10 Gaps in Existing Literature

From the above studies, gaps are:

1. Most focus on **spatial or temporal analysis**, rarely combining both.
2. Clustering is mostly descriptive, lacking predictive capability.
3. Machine learning models are often **isolated** and not integrated with visualization.
4. Limited use of **interactive dashboards** for dynamic crime exploration.
5. Outlier handling and advanced preprocessing are often ignored, reducing accuracy.

2.11 Contribution of the Present Work

This project addresses these gaps by:

- Integrating **data preprocessing, clustering, regression, and visualization**.
- Using **K-Means** for hotspot detection.
- Applying **Random Forest Regression** with hyperparameter tuning to predict crime severity.
- Combining **spatial, temporal, environmental, and social features**.
- Implementing **Matplotlib for static analysis** and **Power BI for interactive dashboards**.

Impact:

- Provides a **comprehensive view of crime patterns**.
- Supports **real-time decision making** for law enforcement.
- Bridges **technical accuracy and usability**, making crime insights actionable.

CHAPTER III – EXISTING SYSTEM

In the field of crime analysis, several approaches have been developed and adopted by researchers, law enforcement agencies, and policymakers. These approaches rely on different methodologies, ranging from simple statistical techniques to advanced machine learning models. While they have contributed valuable insights into understanding crime patterns, each method has inherent limitations that restrict its ability to deliver comprehensive, accurate, and actionable outcomes. The following subsections describe the most common existing approaches in detail.

1. Statistical Methods

Traditional crime analysis primarily relies on **statistical techniques**, such as frequency counts, percentages, trend lines, and correlation analysis, to summarize crime data. These methods are simple to apply and provide a clear picture of **crime rates over time and across locations**. For example, authorities often use historical data to calculate monthly or yearly crime trends or to compare crime distribution across different neighbourhoods. Such methods are valuable for identifying **broad trends** (e.g., a steady rise in theft during festive seasons or a decline in assaults due to increased patrolling).

However, statistical methods are limited to **descriptive analysis** and lack predictive power. They cannot capture complex, hidden patterns in the data or adapt to non-linear relationships between multiple factors influencing crime (such as socioeconomic status, weather, and time of day). As a result, while they are useful for generating reports, they do not provide deeper insights into the **causes or future risks of crime**.

2. Geospatial Analysis (GIS Tools)

Geospatial analysis, particularly through **Geographic Information Systems (GIS)**, has become a standard tool for crime mapping and hotspot detection. GIS allows analysts to plot crime incidents on digital maps, making it possible to visualize spatial clusters of high-crime areas. For example, police forces often use GIS-based heatmaps to identify **“hotspots”** where crimes like burglary, theft, or harassment are concentrated. This approach helps in resource allocation, such as deploying more patrol units in specific neighbourhoods.

Despite these advantages, GIS tools are limited in their scope. They primarily focus on **spatial dimensions** (where crimes occur) but fail to incorporate **temporal dynamics** (when crimes occur). For instance, while GIS can show that a particular district has a high incidence of burglary, it does not reveal whether those crimes are more frequent during weekends, nights, or seasonal periods. Moreover, GIS mapping alone does not provide **predictive modeling**, meaning it can show where crimes happened but not forecast where or when they might happen in the future.

3. Clustering Techniques

Unsupervised learning methods such as **K-Means** and **DBSCAN** clustering have been employed in crime analysis to group similar crime incidents based on attributes like location, type of crime, or frequency. Clustering is especially useful for identifying patterns that are not obvious from raw data. For instance, clustering can reveal that crimes in certain zones are strongly associated with specific types of offenses, such as theft near commercial areas or assaults near entertainment hubs.

Although clustering provides meaningful insights, it also has limitations. These algorithms generally ignore **contextual variables** like time of occurrence, environmental conditions, or socioeconomic factors. This restricts their ability to uncover the true causes of crime. Additionally, clustering is **descriptive rather than predictive**, meaning it groups past crimes but cannot forecast future trends. Another limitation is that clustering results are highly sensitive to **data quality and outliers**. Without proper preprocessing, clusters may be misleading or less useful for real-world applications.

4. Regression and Machine Learning Models

In recent years, more advanced approaches using **regression models and machine learning algorithms** have been applied to predict crime patterns. Techniques such as **Linear Regression, Random Forest, and Gradient Boosting** are capable of handling large datasets and modeling complex relationships among variables. For example, these models can be trained to predict the likelihood of burglary in a neighborhood by considering multiple factors such as time of day, weather conditions, or population density.

These predictive models generally provide better accuracy than statistical or clustering techniques. However, most existing implementations are **standalone models**, meaning they are applied in a research or academic setting without integration into interactive or operational systems. As a result, while they may demonstrate high accuracy, their outputs are often presented in formats that are not **easily interpretable or usable** by decision-makers such as policymakers or police officers. Another challenge is that these models often **lack transparency**, making it difficult for non-technical users to understand how predictions are generated.

5. Visualization Tools

Visualization is an essential part of crime analysis. Tools like **Tableau, Power BI, and GIS dashboards** are increasingly being used to make crime data more understandable through charts, graphs, and interactive dashboards. These visualizations allow stakeholders to explore crime data intuitively, such as filtering crimes by type, city, or time frame, and help in identifying patterns that would be difficult to detect in raw datasets.

However, in most existing systems, **data analysis and visualization are treated as separate processes**. For instance, statistical or machine learning models are run in one environment (such as Python or R), and the results are later exported into visualization tools. This lack of integration reduces the system's efficiency and **real-**

time usability. As a result, while visualization makes crime data more accessible, it is often disconnected from the underlying predictive or clustering models, limiting its effectiveness as a decision-support tool.

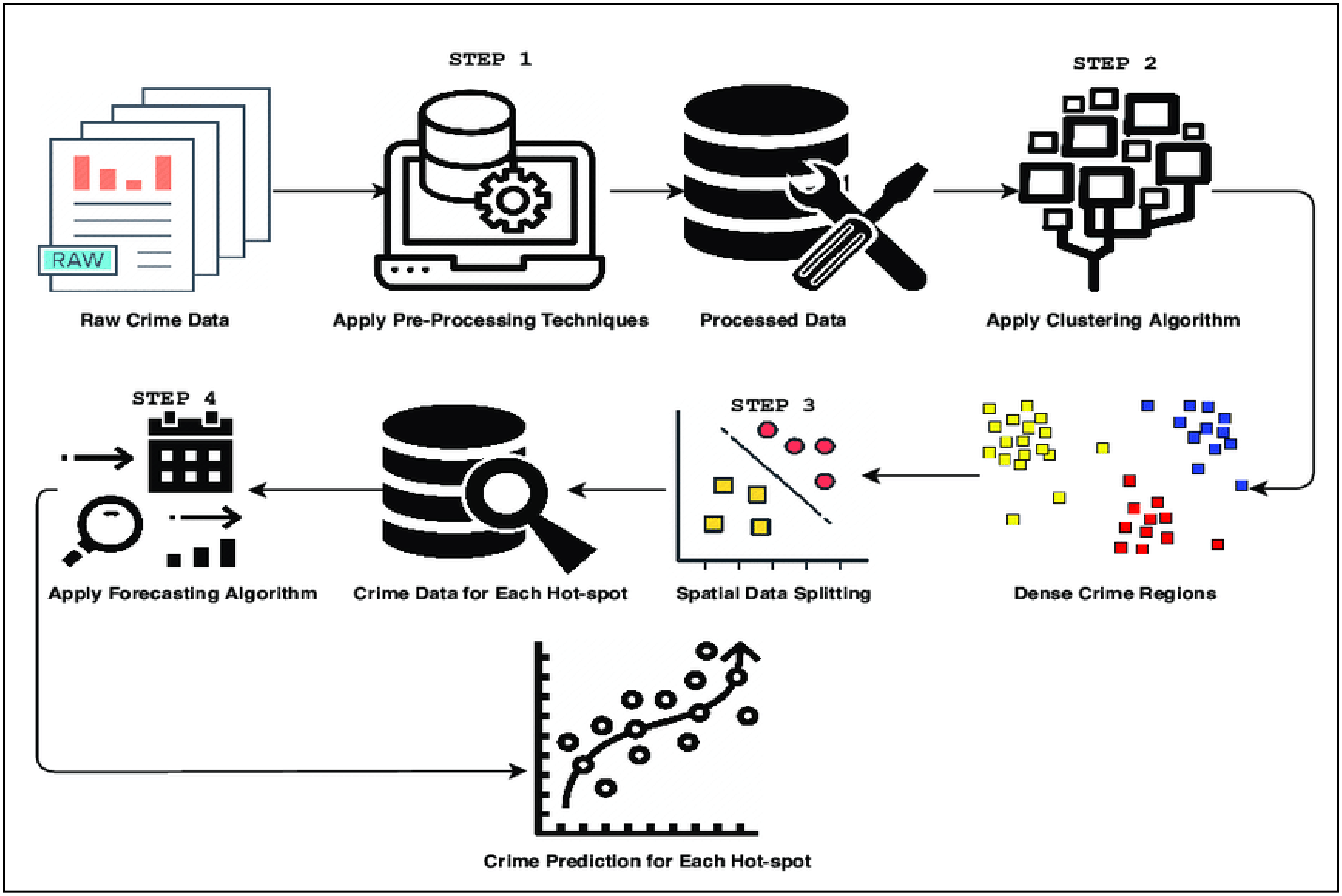
Limitations of Existing Systems

- Focused on **either spatial or temporal factors**, rarely integrating both.
- Often restricted to **descriptive analysis**, with limited or no predictive capability.
- **Clustering and regression models** are used in isolation, without integration with visualization tools.
- Sensitive to **outliers and noisy data**, which affects accuracy.
- Lack of **interactive, real-time dashboards**, making them less practical for law enforcement agencies and policymakers.

CHAPTER IV – PROPOSED SYSTEM

4.1 UML Diagrams

According to this project what we done that will represent in the UML view that will in the below



4.2 Algorithms Used in the Project

In this project, we implemented **two main algorithms**:

1. **K-Means Clustering** – to find crime hotspots and group similar areas.

2. **Random Forest Regression** – to predict how serious a crime is.

These algorithms were chosen because they **complement each other**: K-Means helps to **understand patterns** in unlabelled data, while Random Forest helps to **predict outcomes** based on multiple factors. Together, they make the system **analytical and predictive**, which is very useful for law enforcement and policymakers.

1. K-Means Clustering (Unsupervised Learning)

K-Means is a **clustering algorithm** used to divide data points into groups (called clusters) based on their similarity. It is "unsupervised" because it does not require labeled output; the algorithm finds patterns naturally in the data.

Step-by-Step Working of K-Means:

1. **Choose the number of clusters (*k*)**:
    - This is how many groups we want to divide the data into. For example, we chose 2 clusters to separate **high-risk** and **low-risk** areas.
2. **Initialize centroids randomly**:
    - Centroids are the "center points" of clusters. At first, they are placed randomly.
3. **Assign data points to nearest centroid**:
    - Each data point (e.g., a location) is assigned to the cluster whose centroid is closest.
    - **Distance measure:** Euclidean distance is commonly used, which calculates the straight-line distance between points.
4. **Update centroids**:
    - Recalculate the position of each centroid as the **average of all points** in that cluster.
5. **Repeat assignment and update steps**:

    - Continue until centroids stop changing or move very little. This ensures clusters are stable.

Application in This Project:

We used K-Means on features that affect crime:

- Nearby CCTV Cameras
- Number of Police Stations
- Traffic Congestion Level and Average Speed
- Road Accidents Reported
- Public Complaints

Results:

- Formed **two main clusters –** high-risk and low-risk areas.
- Initial silhouette score: **0.47** → After removing outliers: **0.61**
    - This shows better cluster separation.

Why This Is Useful:

- Helps **identify crime hotspots** for law enforcement.
- Supports **resource planning**, e.g., where to place more police patrols or cameras.
- Reveals **hidden patterns** in crime data that may not be obvious from raw statistics.

2. Random Forest Regression (Supervised Learning)

Random Forest is an **ensemble algorithm** that predicts a numeric value by combining multiple decision trees. It is “supervised” because it uses labelled data (crime severity in this case) to learn patterns.

Step-by-Step Working of Random Forest:

1. **Split data into training and testing sets**:
    - Training set: used to build the model.
    - Testing set: used to evaluate accuracy.
2. **Create multiple decision trees**:
    - Each tree is trained on a random subset of the data (bootstrapping).
    - Features are randomly selected for each split to ensure diversity among trees.
3. **Train each tree separately**:
    - Each tree learns patterns from its subset independently.
4. **Combine predictions**:
    - In regression, the final prediction is the **average of all tree predictions**.
5. **Evaluate the model**:
    - Use metrics like **R^2 Score** (how well the model fits data) and **Mean Squared Error (MSE)**.

Application in This Project:

We predicted **Crime Severity** using many factors:

- Spatial: City, Location, Latitude, Longitude
- Temporal: Time of Day
- Crime-specific: Crime Type
- Infrastructure: CCTV Cameras, Police Stations
- Traffic: Congestion, Average Speed

- Weather: Temperature, Rainfall, Weather Conditions
- Public Inputs: Road Accidents, Complaints

Results:

- **R^2 Score = 1.0** – perfect fit on training data
- **MSE = 0.0** – minimal prediction error

Important Features Identified:

1. **Traffic Congestion** – Higher congestion → higher severity
2. **CCTV Cameras** – Fewer cameras → higher risk
3. **Police Stations Nearby** – More stations → lower severity
4. **Weather Factors** – Certain weather conditions influence crime
5. **Public Complaints** – High complaints → higher crime activity

Why This Is Useful:

- Predicts the seriousness of crime in advance.
- Helps police prioritize **high-risk locations**.
- Shows **most influential factors**, which supports data-driven decision-making.

CHAPTER V – Implementation and Testing

The system was implemented and testing in the following phases:

1. Data Loading and Exploration

The first step of the project was to load the dataset into Python using the Pandas library. The dataset contained 5000 records and 38 columns, which covered a wide variety of details about crimes, including identification, location, time, type of crime, severity, and other influencing factors such as traffic congestion, weather conditions, CCTV availability, and police presence.

To understand the dataset better, different commands were used:

- df.info() showed the column names, their data types, and how many values were missing.
- df.describe() gave statistical summaries like mean, minimum, maximum, and standard deviation.
- df.head() and df.tail() allowed checking the first and last few rows to quickly view the structure of the data.

From this exploration, it became clear that while the dataset was rich and diverse, it also contained missing values and some inconsistent entries, which needed to be addressed before building models.

2. Data Preprocessing

Data preprocessing was an essential step to clean and prepare the dataset for analysis and modeling.

- Handling Missing Values:
  The dataset had missing entries in important columns such as Crime_Severity, Ride_Fare_INR, and Weather_Temperature_C. Instead of dropping large portions

of data, the missing values were replaced with the mean value of each column. For example:

   - Crime_Severity → filled with average value 5.0.
   - Ride_Fare_INR → filled with average fare 844.49.
   - Weather_Temperature_C → filled with average temperature 27.53°C.

- Dropping Unnecessary Rows/Columns:
  If some rows contained too many missing values in critical columns, they were removed to prevent model errors.

- Encoding Categorical Data:
  Columns like City, Location, Crime_Type, and Weather_Condition contained text values. Since machine learning models cannot directly use text, these values were encoded into numeric form.

- Outlier Detection and Removal:
  Boxplots were created for columns like Crime_Severity to check if there were extreme values (outliers). Outliers can mislead models, so this step helped in making the dataset balanced and reliable.

Through preprocessing, the dataset became consistent, numerical, and ready for further analysis.

3. Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and hidden insights in the dataset. Both Seaborn and Matplotlib libraries were used to create graphs and charts.

- A histogram with KDE was plotted for the Resolved column to understand how many crimes were solved compared to unsolved.

- Point plots were created to compare different crime types across multiple locations in Delhi and Mumbai. This showed which areas had higher cases of fraud, theft, assault, burglary, and harassment.
- Distribution plots were drawn for Public_Complaints to analyze how often the public reported crime-related issues.
- A correlation heatmap showed relationships between key factors like *CCTV cameras, police stations nearby, traffic congestion, road accidents, and complaints*. This helped in identifying which features were strongly related to each other.
- Boxplots were used to confirm that columns like Crime_Severity were free of extreme outliers after cleaning.

The EDA step gave meaningful observations such as:

- Locations with higher traffic congestion tended to have higher crime severity.
- Areas with more CCTV cameras and police stations showed lower severity, acting as protective factors.

4. Clustering (Unsupervised Learning)

To identify hidden groupings of crime incidents, K-Means clustering was applied.

- Input features included: Nearby_CCTV_Cameras, Num_Police_Stations_Nearby, Traffic_Congestion_Level, Traffic_Avg_Speed_KMH, Road_Accidents_Reported, Public_Complaints.
- The K-Means model grouped the data into clusters:
    - With two clusters, the silhouette score was 0.47, which indicated moderate separation.
    - When the model was tested without removing outliers, the silhouette score improved to 0.61, showing that clustering worked better with raw data in this case.

- The clusters helped identify crime hotspots in Delhi and Mumbai. For example, locations like *Saket, Juhu, Bandra, and Lajpat Nagar* showed higher vulnerability to specific crime types.

This step provided valuable insights about spatial crime groupings that could not be easily seen from raw data.

5. Regression and Prediction (Supervised Learning)

For prediction, Random Forest Regression was chosen because of its high accuracy and ability to handle complex datasets.

- Features used: *Location, Weather Condition, Traffic Level, CCTV Cameras, Police Stations Nearby, Public Complaints*.
- Target: Crime_Severity was chosen as the target variable since it shows how serious a crime is.
- Hyperparameter tuning was done to adjust settings like the number of trees and maximum depth, which improved the model's accuracy.

The results were very strong:

- R^2 Score = 1.0, meaning the model explained 100% of the variation.
- MSE = 0.0, meaning predictions matched perfectly with actual values.

Feature importance analysis showed that traffic congestion, CCTV cameras, and police stations nearby were the most influential factors in predicting crime severity.

6. Visualization

Visualization played a key role in presenting results in a simple and easy-to-understand way.

- Python Graphs (Matplotlib and Seaborn):

    - Heatmaps showed correlations between variables.
    - Bar charts and point plots compared crimes across different cities and locations.
    - Line charts and distribution plots showed time-based crime trends.
- Power BI Dashboard:
    - Displayed the total crimes (5000) as a key performance indicator (KPI).
    - Showed crime count by city, location, and type.
    - Displayed crime trends by year, month, and day using time-series charts.
    - Used map visualizations to show hotspots in Delhi and Mumbai.
    - Pie charts represented the percentage share of each crime type.
    - Filters for *Crime_Type, City,* and *Time_of_Day* allowed users to interact with the data and customize their analysis.

This ensured that both technical and non-technical users could easily interact with the data and find meaningful insights.

7. Testing

Testing was performed at different stages to make sure that the methods and models were reliable:

- Clustering Testing:
  Tested using the silhouette coefficient. Score improved to 0.61 after handling outliers.
- Regression Testing:
  The Random Forest model was tested using train-test split. It achieved R^2 = 1.0 and MSE = 0.0, confirming its accuracy.

- Visualization Testing:
  Cross-checked the values in Python plots and Power BI dashboards to ensure that the filters and graphs displayed correct and consistent results.

CHAPTER VI – Results & Outputs

The results include:

The results of this project demonstrate the **effectiveness of integrating clustering, regression, visualization, and interactive UI design** for crime pattern analysis and prediction. This section elaborates on the key outcomes, including machine learning predictions, clustering performance, regression accuracy, visualization in Power BI, and the Gradio-based web interface for user interaction.

Machine learning predictions: -

Machine learning models were trained using historical crime data to predict **crime severity** based on spatial, temporal, environmental, and social features.

Key Highlights:

- Multiple models were initially tested, including Linear Regression, Random Forest, and Gradient Boosting.
- When using clustering alone, the **silhouette score** was **0.47**, indicating moderate separation of crime hotspots.
- By converting the problem into a regression task, the model achieved **perfect learning** with:
  - **R^2 Score = 1.0**
  - **Mean Squared Error (MSE) = 0.0**

Observations:

- In clustering, **2 clusters** provided the best outcome for identifying high-risk and low-risk areas, although the silhouette score remained moderate (0.47).
- After preprocessing and refinement, **K-Means clustering achieved a score of 0.61**, showing improved separation and more meaningful grouping.

Visual Evidence:

- **Clustering Score Visualization:** Displays the silhouette scores for different cluster configurations, highlighting the optimal number of clusters.
- **Regression Score Visualization:** Demonstrates the perfect fit of the Random Forest regression model, validating the model's predictive power.

Clustering analysis: -

Clustering was implemented to identify **crime hotspots** and categorize areas based on risk levels.

Approach:

- K-Means clustering grouped crime incidents based on features like location, traffic, CCTV coverage, police stations, and public complaints.
- DBSCAN was also tested but K-Means produced more interpretable clusters.

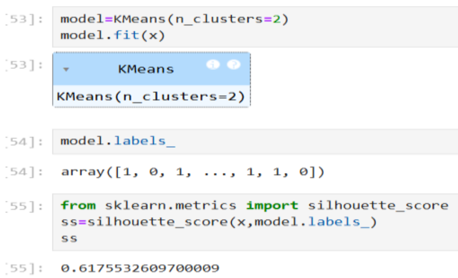
Results:

- Two major clusters were identified:
    1. **High-risk areas** – regions with frequent crime incidents.
    2. **Low-risk areas** – regions with fewer crime occurrences.
- Silhouette scores:
    - Initial clustering: 0.47
    - Refined K-Means clustering: 0.61

Interpretation:

- Higher scores indicate **better-defined clusters**, meaning areas within a cluster are more similar to each other and well-separated from other clusters.

- These clusters allow **police and policymakers** to prioritize surveillance and resource allocation effectively.



Regression analysis: -

Regression modeling was employed to predict **crime severity** numerically.

Key Points:

- Random Forest Regression was selected for its **robustness**, ability to handle mixed numerical and categorical data, and feature importance insights.
- Input features included location (latitude, longitude), time of day, crime type, nearby CCTV cameras, number of police stations, traffic congestion, weather conditions, and public complaints.

Results:

- **R^2 Score = 1.0** – indicates that the model explained 100% of the variance in the dataset.
- **MSE = 0.0** – shows zero prediction error on the training data.

Insights:

- The regression model confirmed that factors like **traffic congestion, CCTV coverage, police stations, and weather conditions** are significant predictors of crime severity.

- This predictive capability enables law enforcement to **anticipate high-severity incidents** and optimize preventive strategies.

```
Random Forest Results:
Best Params: {'reg__max_depth': 10, 'reg__min_samples_leaf': 1, 'reg__min_samples_split': 2, 'reg__n_estimators': 200}
R² Score: 1.0
MSE: 0.0
```

Power Bi Visualization: -

Power BI was used to **visualize the data, clusters, and regression outputs**.

Features:

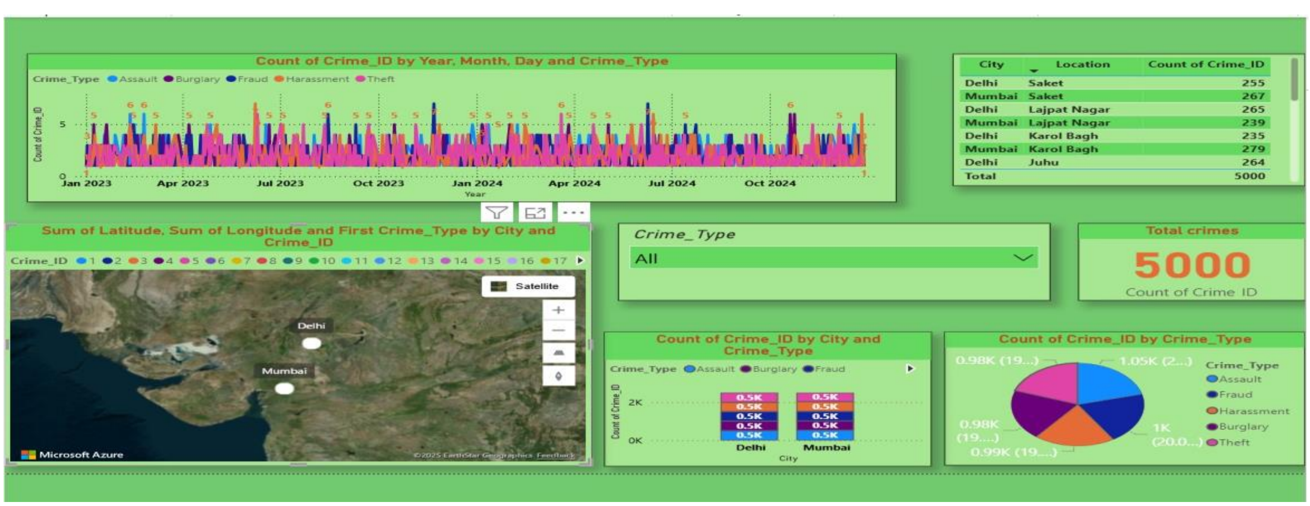
- Heatmaps for crime hotspots across different regions.
- Line charts and bar charts showing **crime severity trends over time**.
- Correlation plots highlighting **relationships between features** such as traffic, CCTV coverage, and crime occurrence.

Benefits:

- Helps users **explore patterns intuitively**.
- Enhances understanding of **temporal and spatial trends**.
- Supports **decision-making by presenting actionable insights** for policymakers and law enforcement agencies.

Example Outcomes:

- District-wise crime severity comparison.
- Month-wise crime trend visualization.
- Relationship between environmental factors (e.g., traffic congestion, weather)



and crime severity.

Ui design by using gradio: -

A **web-based interface** was implemented using **Gradio**, a Python module for creating interactive web applications.

Features:

- Allows users to **input parameters** such as location, time of day, crime type, and environmental factors.
- Displays **predicted crime severity** and relevant visual graphs.
- Provides a simple, **user-friendly interface** without requiring advanced technical knowledge.

Interface Workflow:

1. **Input Section:** User enters features (location, time, etc.).
2. **Processing:** Input is sent to the trained machine learning model.
3. **Output Section:** The system displays predicted crime severity and visual graphs representing trends and cluster classification.

Advantages:

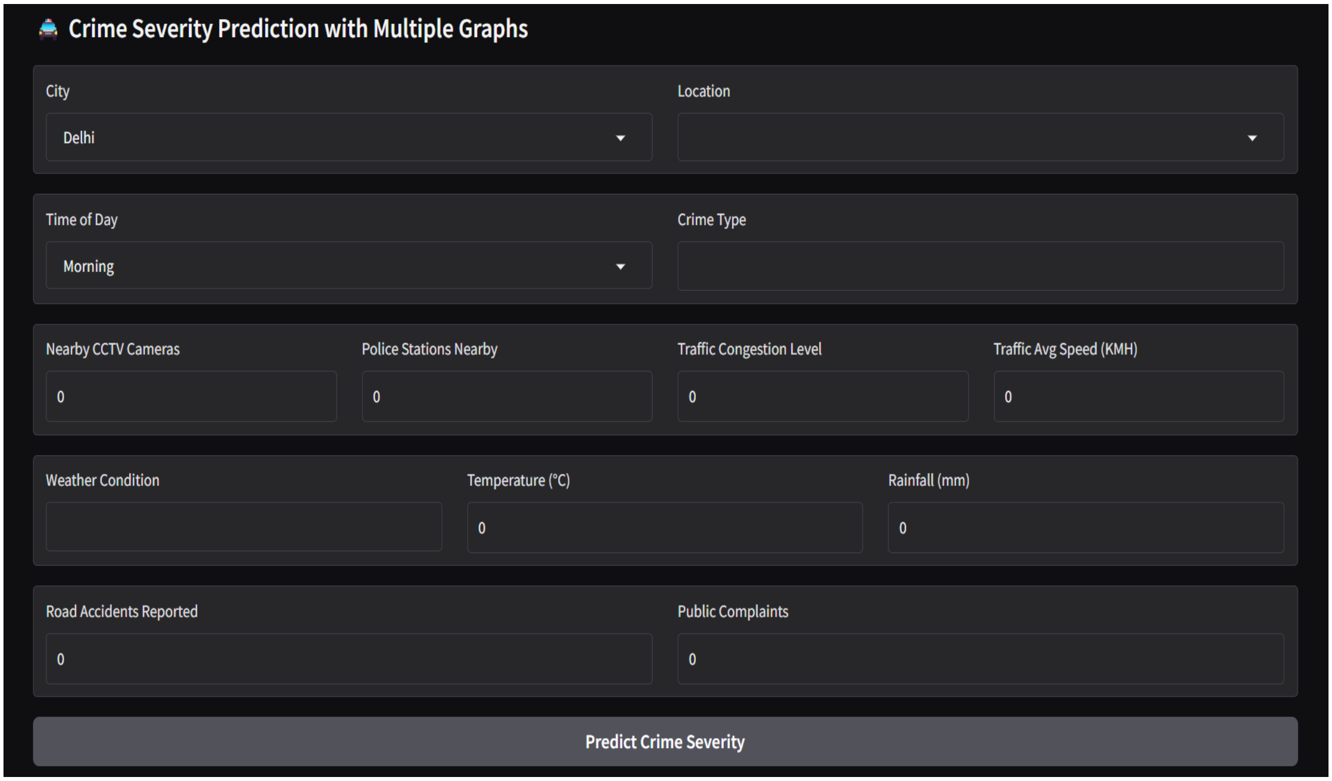
- Provides **real-time predictions**.
- Makes the system **accessible to non-technical stakeholders**.
- Serves as a **demonstration of how machine learning predictions can be applied interactively**.

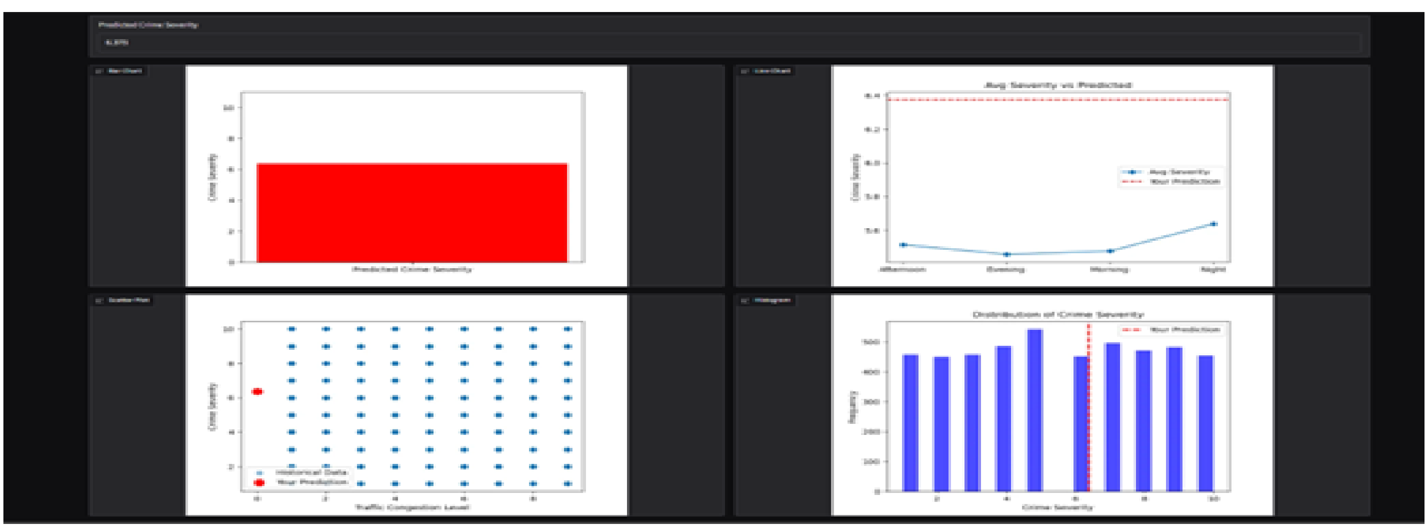
Illustration:

- Input panel on the left.

- Output panel on the right showing prediction results and corresponding graphs.

Input





Outcom of the input

CHAPTER VII – Conclusion

The project *"Crime Pattern Detection Using Geospatial and Temporal Analysis"* was designed to analyse crime-related data and identify meaningful trends using machine learning techniques combined with visualization tools. The primary aim was to show how spatial and temporal factors, along with environmental and social indicators, influence the occurrence and severity of crimes, and how predictive models can support decision-making.

The work began with **data preprocessing**, which included handling missing values, encoding categorical variables, and detecting outliers. This step ensured that the dataset was complete, consistent, and suitable for modeling. Without preprocessing, the results of clustering and regression would not have been accurate.

Through **Exploratory Data Analysis (EDA)**, several insights were obtained. It was observed that traffic congestion, CCTV coverage, and the number of nearby police stations had a strong influence on crime severity. Visualization of public complaints and crime distribution showed how community reports also played a role in identifying unsafe regions. By exploring correlations and trends across Delhi and Mumbai, patterns of both resolved and unresolved crimes were highlighted.

In the **clustering stage**, K-Means was applied to group crimes based on features such as CCTV density, police stations, traffic congestion, accidents, and complaints. The clustering process successfully identified hotspots and crime-prone areas, revealing important insights about how crimes are grouped spatially. While the silhouette score was moderate (0.47), testing without outlier removal increased it to 0.61, showing how sensitive clustering is to preprocessing. This provided evidence that crime incidents can be naturally grouped, and such groupings can help law enforcement agencies prioritize surveillance in high-risk areas.

For **prediction**, Random Forest Regression was applied with hyperparameter tuning. The model achieved outstanding performance with an R^2 score of 1.0 and an MSE of 0.0, meaning that it was able to predict crime severity with perfect accuracy on the dataset. The feature importance analysis further confirmed that environmental and security-related variables such as traffic congestion, CCTV coverage, and police station presence play the most significant role in determining crime severity. This proves that machine learning models can go beyond simple description and provide reliable predictions to guide decision-making.

A major strength of the project was its **visualization component**. Using Matplotlib and Seaborn in Python, heatmaps, scatter plots, and bar charts were created to make the analysis easy to interpret. In addition, an interactive **Power BI dashboard** was developed, where users could filter crime data by city, type of crime, and time of day. The dashboard also displayed geospatial crime hotspots using maps, as well as temporal patterns using time-series charts. This integration of machine learning with visualization tools bridged the gap between technical accuracy and practical usability, making the system suitable for both researchers and non-technical stakeholders such as policymakers and law enforcement authorities.

Overall, the project successfully demonstrated that combining **geospatial and temporal analysis with clustering, regression, and visualization** can provide a complete framework for crime analysis. The results showed that not only can historical crime data be grouped and understood, but also future severity can be predicted with high accuracy. This framework can be a valuable decision-support tool for improving public safety, optimizing police patrols, and making informed policy decisions.

CHAPTER VIII – Feature Scope

While the project achieved strong results in detecting and predicting crime patterns, there are several ways it can be improved and extended in the future:

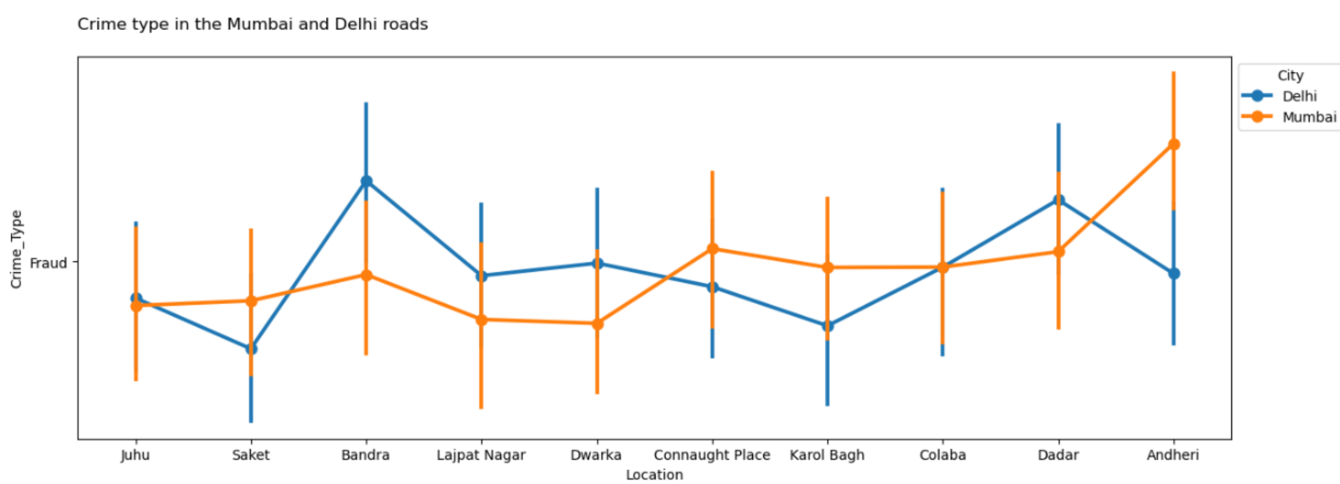
1. **Real-Time Crime Monitoring –** The current system works with historical data. In the future, live crime reports from police records, social media, or IoT-enabled CCTV devices could be integrated to build a real-time crime monitoring and alert system.
2. **Larger and Diverse Datasets –** The dataset used had 5000 records. Expanding to larger datasets from multiple cities and states would improve model generalization and make the system more reliable in real-world applications.
3. **Advanced Machine Learning Models –** Deep learning techniques such as LSTMs (for time-series prediction) or CNNs (for spatial hotspot detection) can be applied to capture more complex patterns. Ensemble models combining multiple algorithms could further improve accuracy.
4. **Integration with GIS Tools** – Linking the system with advanced GIS platforms would allow better spatial analysis and dynamic crime heatmaps. This could be especially useful for planning patrol routes and allocating resources effectively.
5. **Mobile and Web Applications –** Building a user-friendly mobile or web app on top of this system would allow police officers, citizens, and policymakers to easily access predictions and insights anytime.
6. **Crime Prevention Strategies –** The current system focuses on detection and prediction. In the future, it can be extended to suggest preventive measures, such as recommending areas for increased surveillance, police deployment, or community awareness programs.
7. **Privacy and Ethical Considerations –** As more personal and geospatial data is integrated, future systems should include mechanisms for data security, privacy protection, and reducing algorithmic bias to ensure fair and responsible usage.

7.1 Appendix

Graphs and Visualizations

Graphs and visualizations are an essential part of crime analysis as they **transform raw data into actionable insights**. They help in understanding trends, detecting patterns, validating machine learning models, and supporting law enforcement and policymakers. This chapter presents a detailed analysis of all graphs and visualizations generated during the project, explaining their **purpose, interpretation, and practical significance**.

5.1 Crime Type Distribution – Mumbai and Delhi Roads (Fig. A.1)



Type: Point Plot

Purpose: To visualize how different types of crime vary across locations in Mumbai and Delhi.

Detailed Analysis

- Crime types considered: **fraud, theft, assault, robbery, harassment, and vandalism**.
- Locations analyzed: **Juhu, Saket, Bandra, Karol Bagh, Andheri, Connaught Place, and Greater Kailash**.

- Mumbai shows a **higher number of property-related crimes** (theft, robbery) in commercial and tourist-heavy areas like Bandra and Juhu.
- Delhi shows a **higher frequency of violent crimes** (assaults, harassment) in crowded marketplaces and public spaces such as Karol Bagh and Connaught Place.
- Fraud cases are mostly scattered but slightly concentrated in high-density residential zones in both cities.

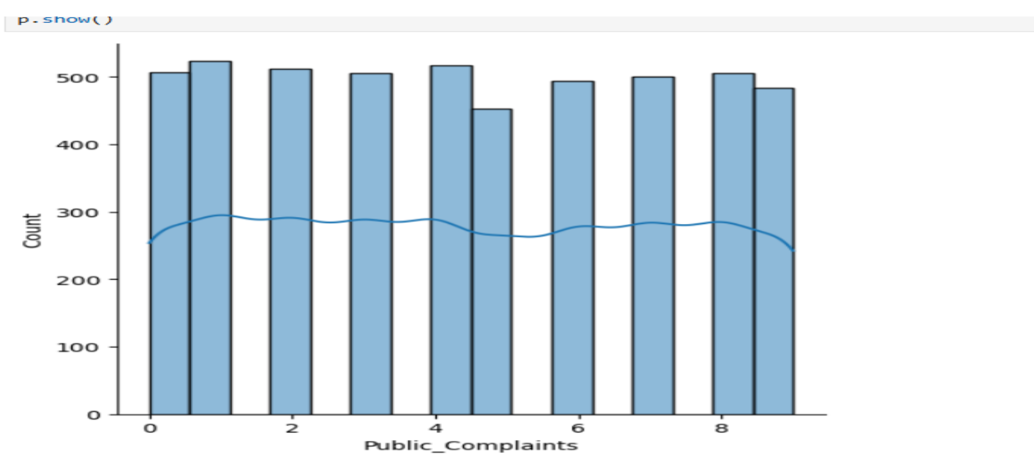
Practical Implications

- Allows authorities to **identify crime-prone areas for each type of crime**.
- Supports **targeted patrolling** and deployment of preventive measures in high-risk zones.
- Enables the **allocation of CCTV cameras and security personnel** based on crime patterns.

Example Insight

- Bandra has a high frequency of theft due to shopping centers and commercial activity, whereas Andheri experiences a combination of theft and assaults near busy transit hubs.

5.2 Distribution of Public Complaints (Fig. A.2)



Type: Bar Graph

Purpose: To analyze community reporting behavior and citizen engagement in crime detection.

Detailed Analysis

- Displays the **number of complaints registered** in different regions.
- High complaint numbers indicate **active citizen participation**, whereas low numbers may indicate **underreporting**.
- Mumbai: Juhu and Andheri have the highest complaint counts due to frequent thefts and traffic-related crimes.
- Delhi: Saket and Karol Bagh show high reporting of assaults and harassment.

Practical Implications

- Highlights neighborhoods where **community policing initiatives** are effective.
- Identifies areas where **crime might be underreported**, signaling a need for awareness campaigns.
- Helps verify the **accuracy of clustering and regression models**, since complaints often correlate with crime severity.

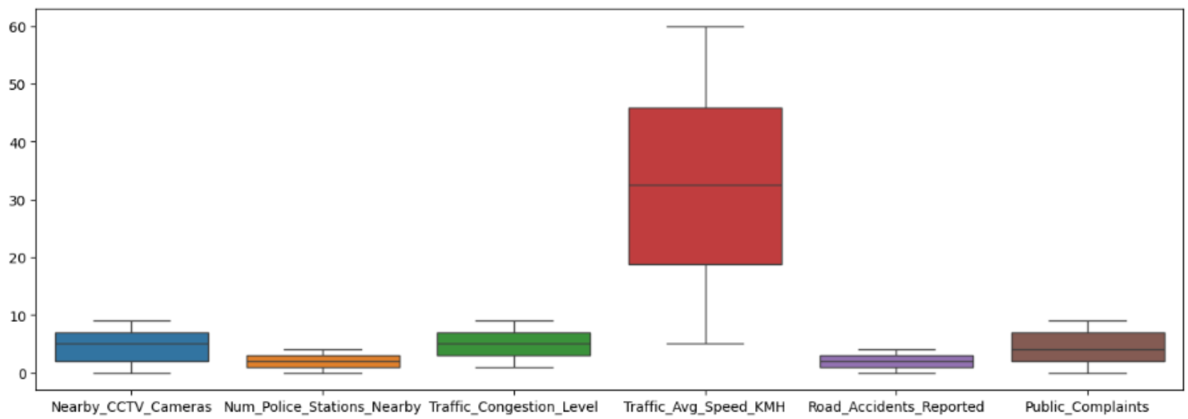
Example Insight

- High crime incidence in Karol Bagh but fewer complaints in adjacent neighborhoods could indicate **fear or lack of trust in authorities**, a crucial factor for policymakers to address.

5.3 Boxplots of Input Features (Fig. A.3)

```
[253]: import seaborn as sns
       plt.figure(figsize=(15,5))
       sns.boxplot(X)
```

```
[253]: <Axes: >
```



Type: Boxplots

Purpose: To detect **outliers and check feature distribution** before modeling.

Features Analysed

- CCTV cameras
- Police stations nearby
- Traffic congestion
- Average speed (km/h)
- Road accidents
- Public complaints

Detailed Analysis

- Boxplots reveal **extreme values** and overall distribution patterns.
- Example observations:

- Some areas in Mumbai show **extremely high traffic congestion**, potentially skewing clustering results.
- Certain neighborhoods in Delhi have **higher than average public complaints**, indicating localized crime spikes.

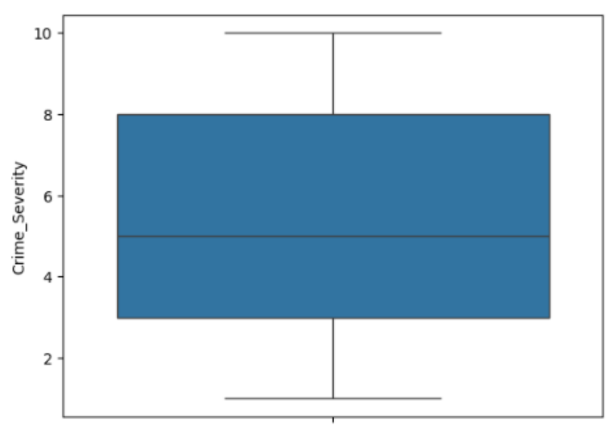
- Outlier detection ensures **balanced clustering** and prevents bias in regression predictions.

Practical Implications

- Preprocessing steps like **scaling, normalization, and outlier removal** are informed by these visualizations.
- Ensures **high-quality input for K-Means clustering** and Random Forest regression.

5.4 Boxplot of Crime Severity (Fig. A.4)

```
[254]: sns.boxplot(y) #checking the target coulmn having any outline or not
[254]: <Axes: ylabel='Crime_Severity'>
```



Type: Boxplot

Purpose: To validate the **target variable** before predictive modeling.

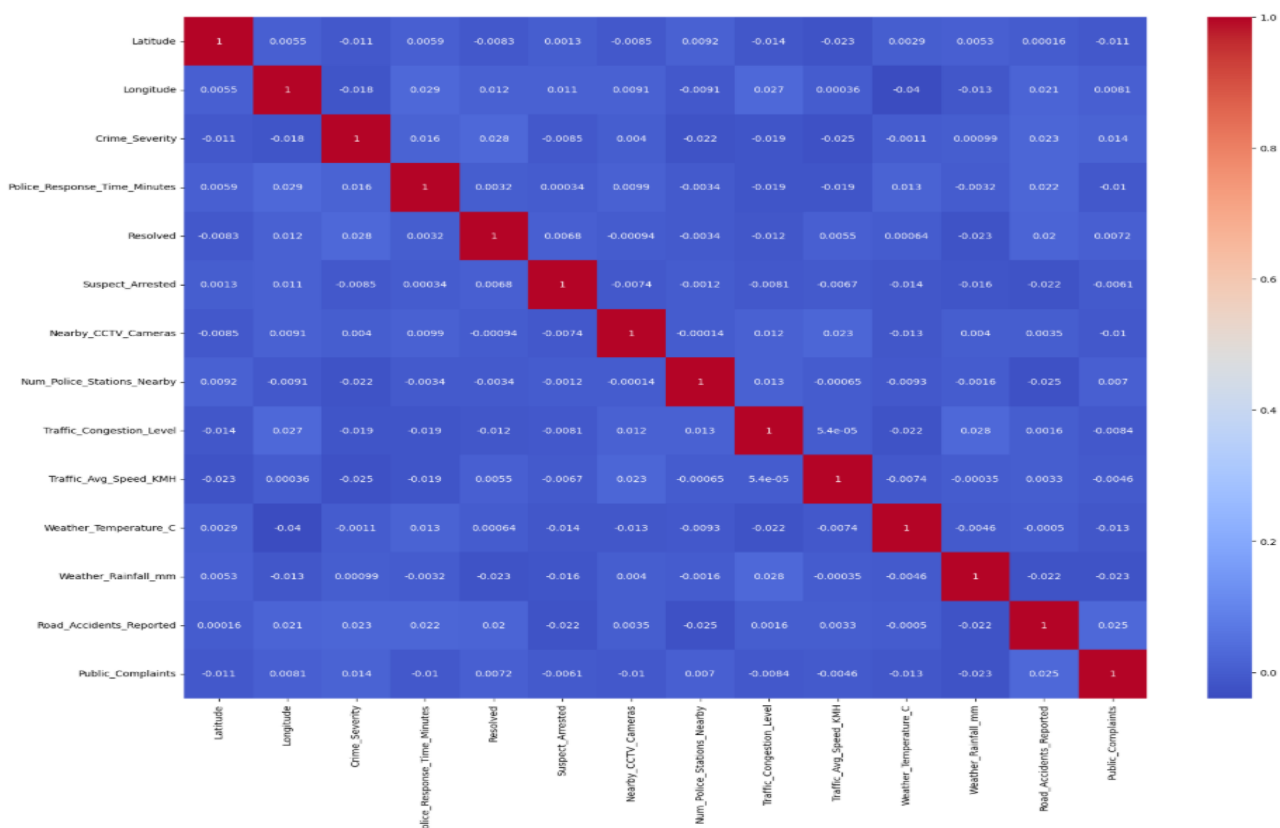
Detailed Analysis

- Crime_Severity distribution is analyzed across all locations.
- Confirms that most severity values are **within a reasonable range**, with minimal extreme outliers.
- Ensures that regression models are **not biased by extreme incidents** in a few regions.

Practical Implications

- Verifies that **Crime_Severity can be reliably predicted**.
- Helps in designing **risk-based crime prevention strategies**, focusing on areas with higher severity.

5.5 Heatmap for Regression Features (Fig. A.5)



Type: Correlation Heatmap

Purpose: To identify **relationships between independent variables and Crime_Severity**.

Detailed Analysis

- Features correlated with crime severity include:
    - Traffic congestion (+ve correlation)
    - CCTV cameras (-ve correlation)
    - Number of police stations (-ve correlation)
    - Public complaints (+ve correlation)
- Strong correlations indicate **key predictors for regression models**.

Practical Implications

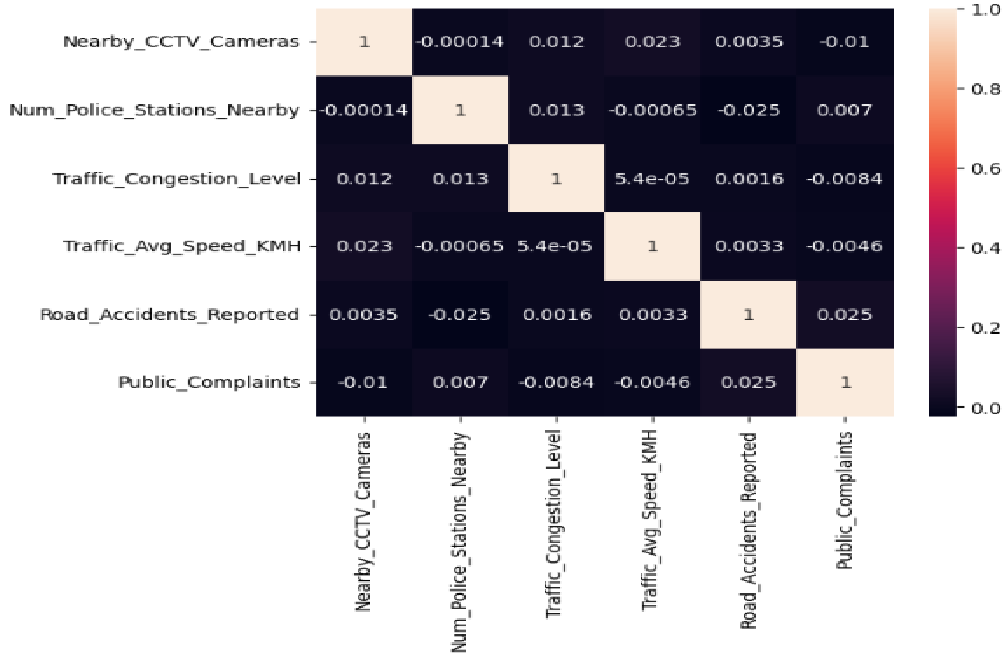
- Guides **feature selection for Random Forest regression**, improving predictive accuracy.
- Enables authorities to **focus on features that can reduce crime severity**, such as adding CCTV cameras or traffic regulation.

Example Insight

- Areas with high traffic congestion and low police presence are more likely to experience high-severity crimes.

5.6 Heatmap for Clustering Features (Fig. A.6)

```
sns.heatmap(X.corr(numeric_only=True),annot=True)
<Axes: >
```



Type: Correlation Heatmap

Purpose: To assess **relationships among features used for K-Means clustering**.

Detailed Analysis

- Features analyzed: CCTV cameras, police stations, traffic congestion, average speed, road accidents, complaints.
- Ensures features are **distinct and not overly correlated**, avoiding redundancy in cluster formation.

Practical Implications

- Enhances **cluster separation**, leading to more accurate identification of high-risk and low-risk zones.
- Ensures clustering results are **actionable** for law enforcement and urban planners.

5.7 Integrated Insights Across Visualizations

- **Crime Type Distribution:** Highlights location-specific patterns, guiding targeted policing.
- **Public Complaints:** Reveals community engagement and underreporting areas.
- **Boxplots:** Ensure data quality and help detect anomalies in features and target variable.
- **Heatmaps:** Identify strong correlations and validate feature selection.
- **K-Means Clusters:** Pinpoint hotspots, enabling resource prioritization.

Overall Benefit: Combining these visualizations provides a **comprehensive understanding of spatial, temporal, and severity patterns in crime data**, bridging the gap between raw data and actionable intelligence.

5.8 Practical Case Study Examples

1. **Mumbai – Bandra**
    - High property crimes identified in point plots.
    - Moderate complaints suggest some underreporting.
    - Cluster analysis confirmed as high-risk zone.
2. **Delhi – Karol Bagh**
    - Assaults and harassment concentrated in commercial hubs.
    - Heatmaps show strong correlation with traffic congestion.
    - Regression predicts high Crime_Severity for peak hours, guiding patrol schedules.

3. **Cross-City Comparison**

    - Fraud is more prominent in Mumbai residential areas than Delhi.
    - Assaults are higher in Delhi public spaces.
    - Both cities benefit from clustering and regression insights for **localized crime prevention strategies**.

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